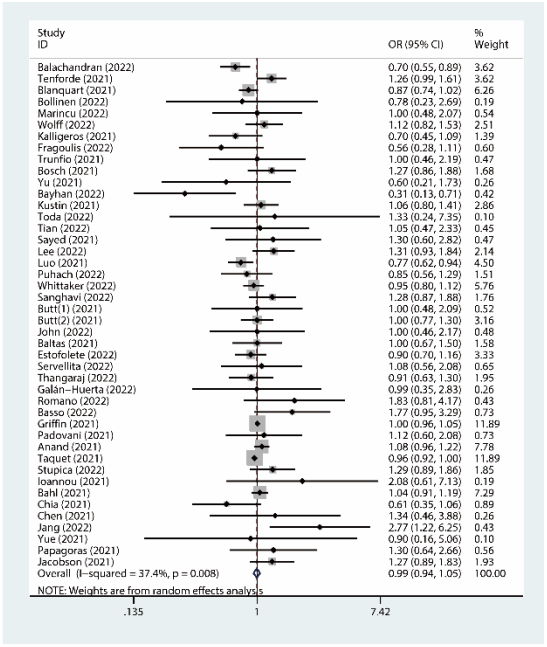
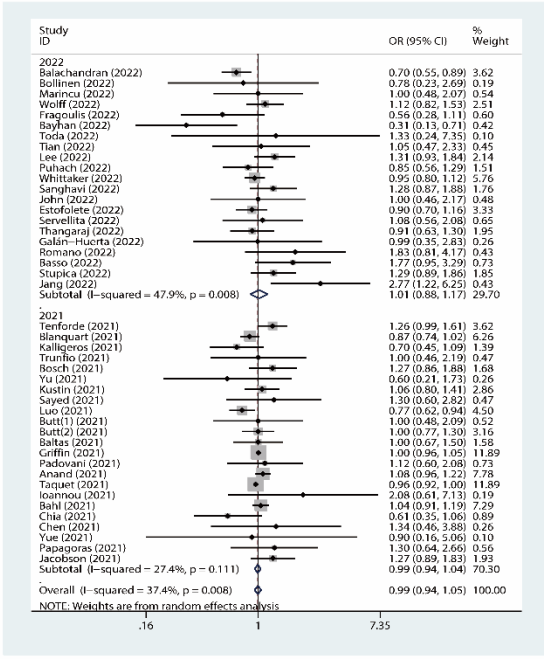
Figure S1. Forest plots and funnel plot of age of COVID-19 patients

a



b



c

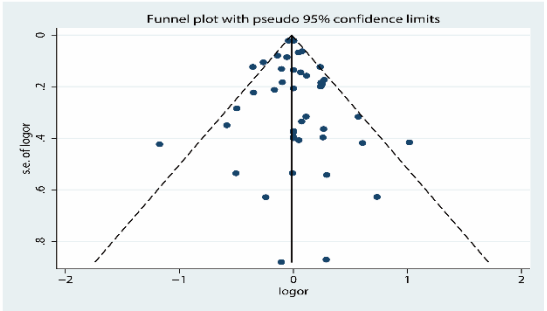
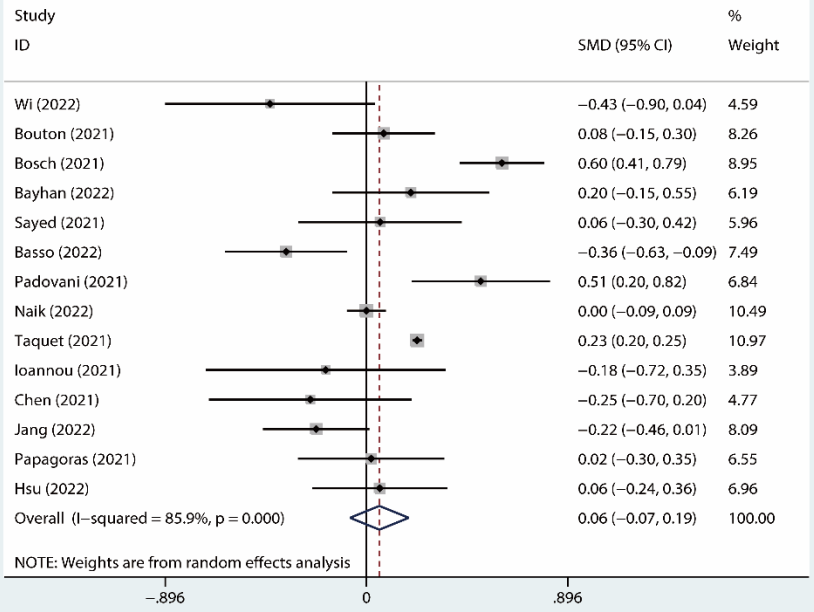


Figure S2. Forest plots and subgroup analysis of age of COVID-19 patients

a



b

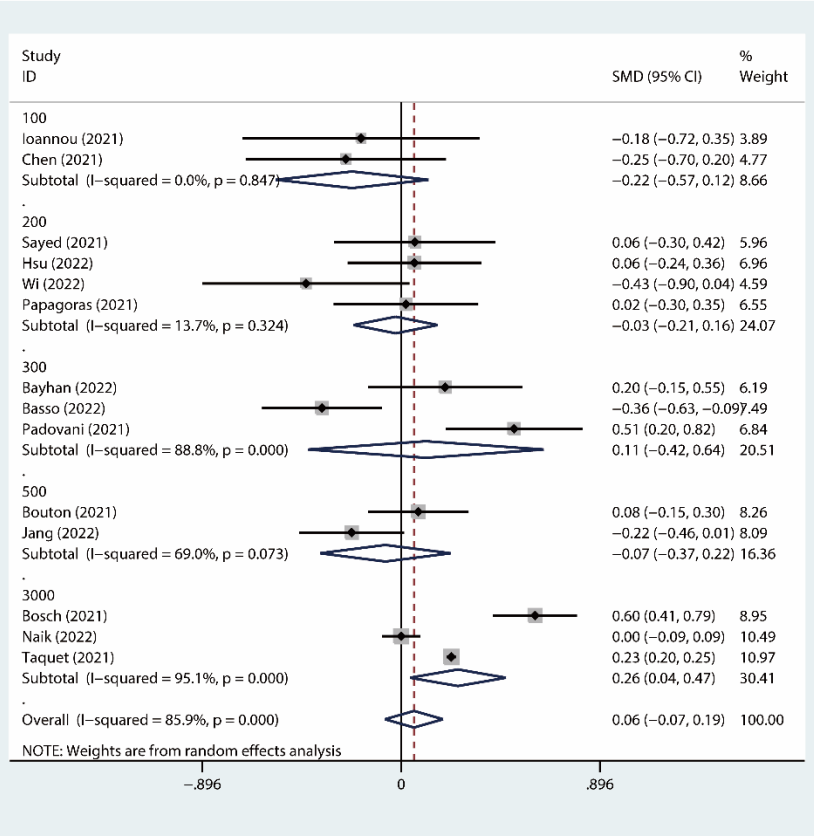


Figure S3. Forest plots and subgroup analysis of disease severity of COVID-19 patients

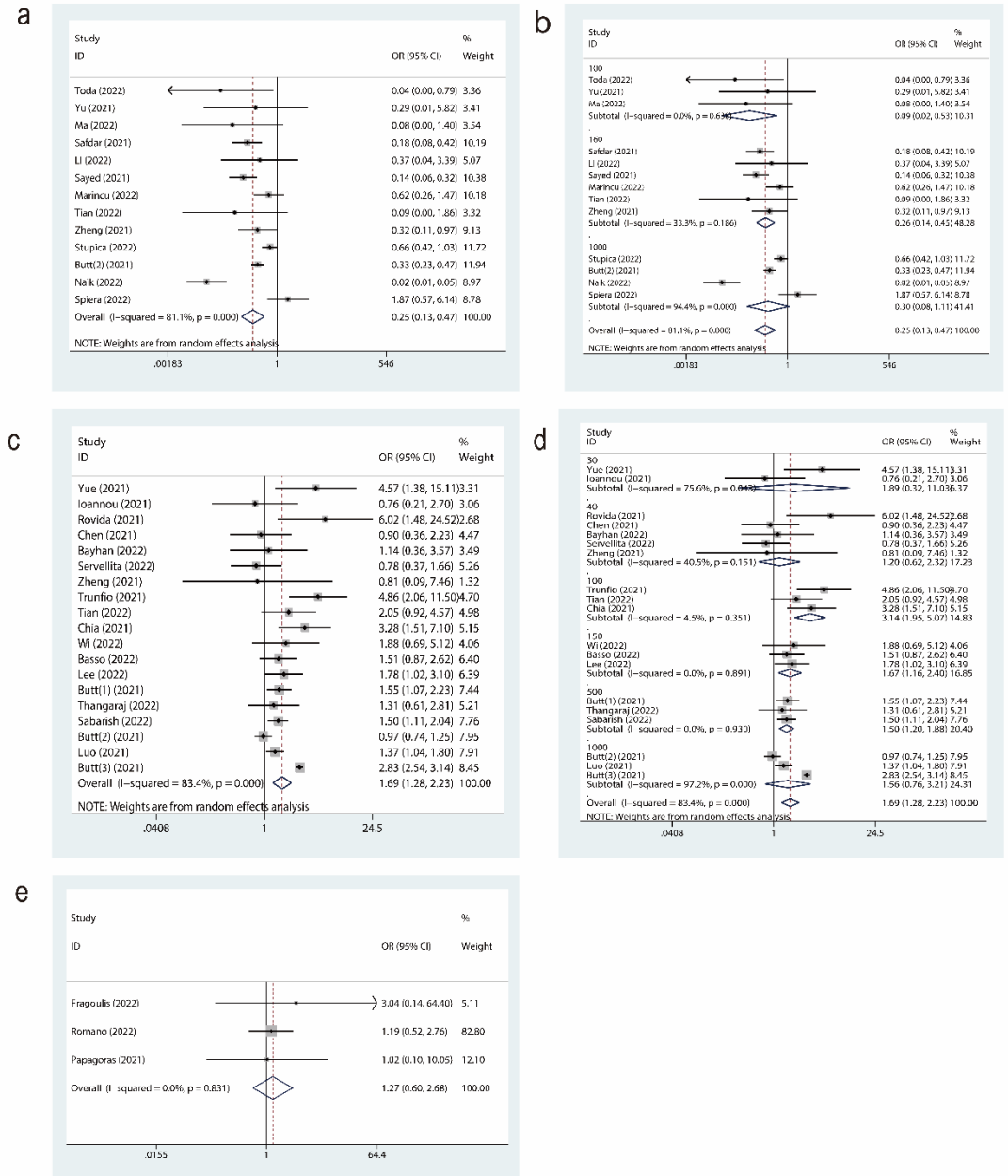


Figure S4. Forest plots of epidemiological and personal history of COVID-19 patients

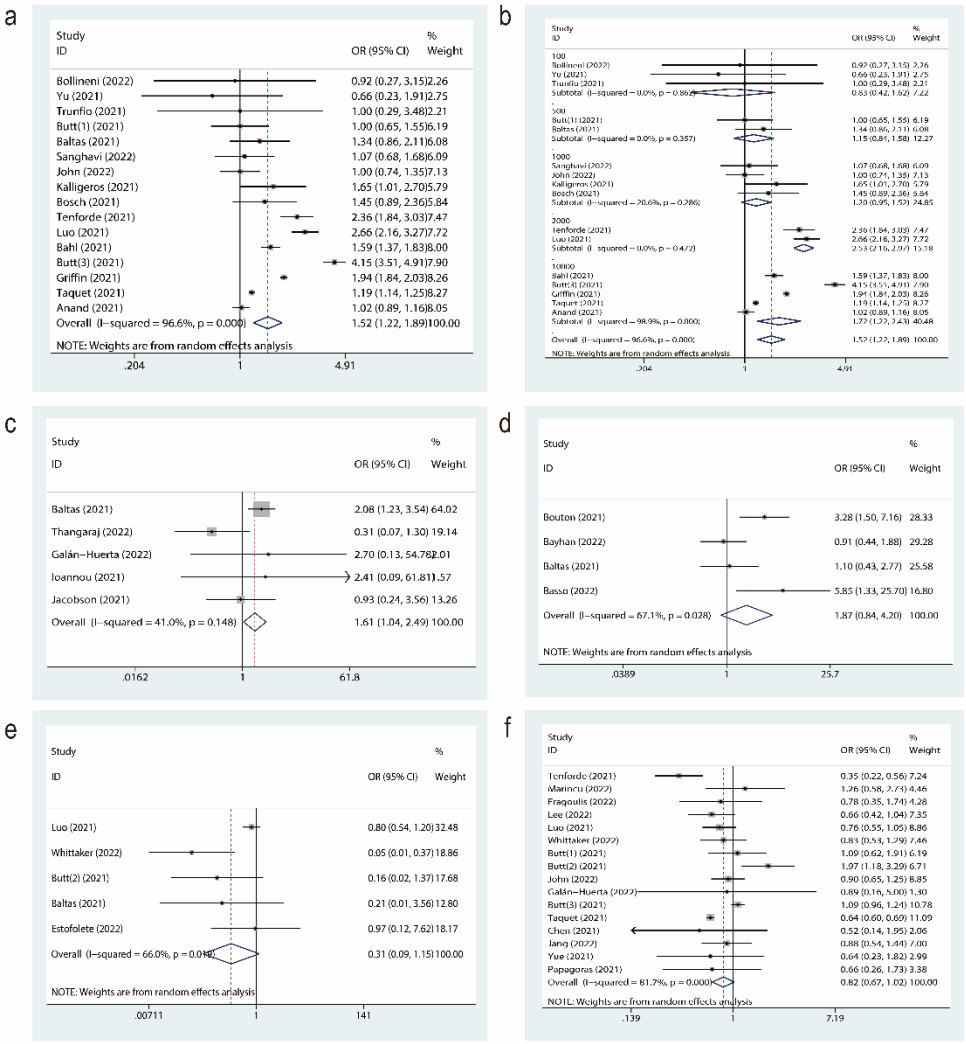


Figure S5. Presence of comorbidity related forest (a), diabetes related forest (b), subgroup analysis of diabetes related forest (c), hyperlipidemia related forest (d), obesity related forest (e), and subgroup analysis of obesity related forest (f)

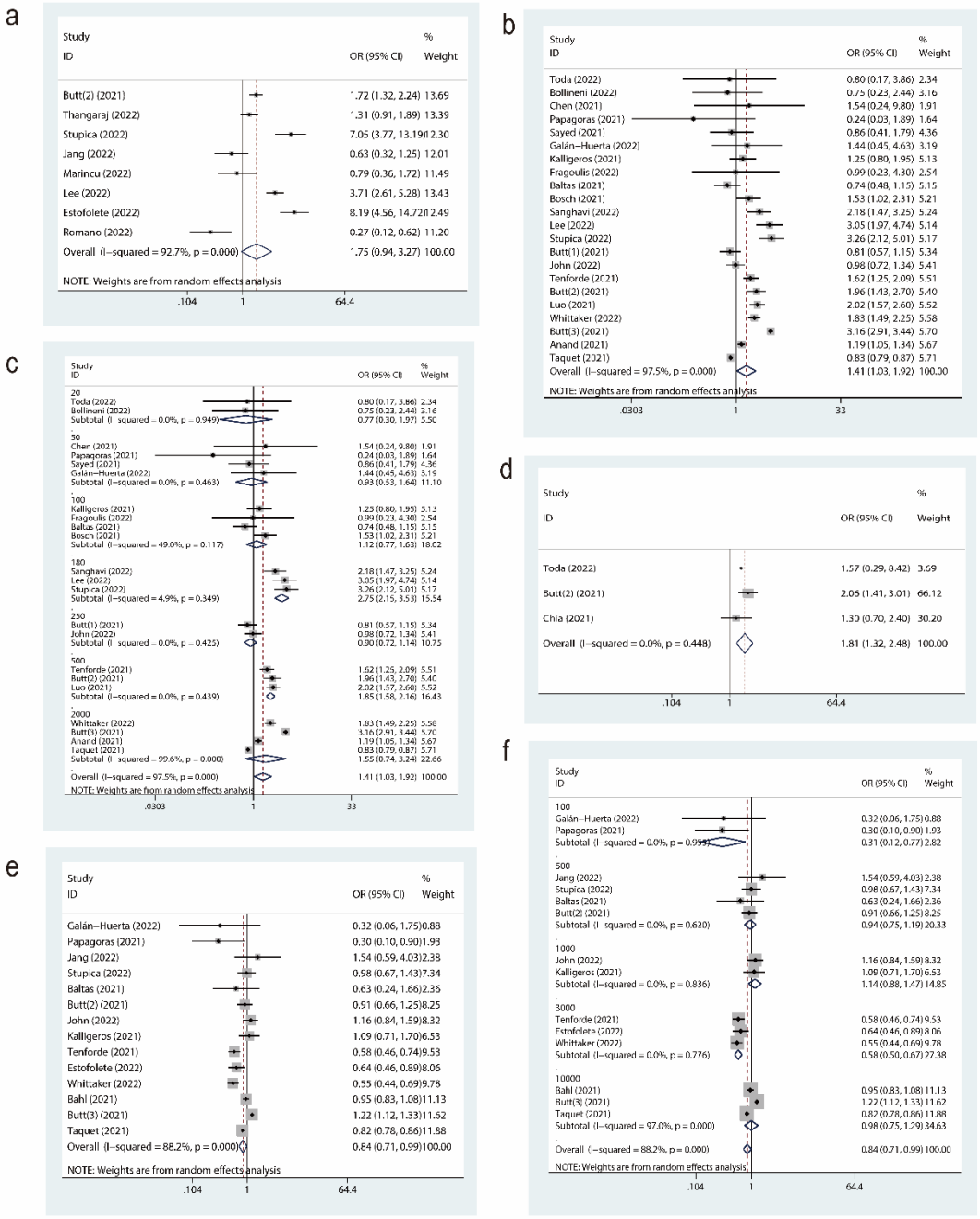


Figure S6. Kidney disease related forest (a), immunocompromised patients related forest (b), organ transplants related forest (c), cardiovascular related forest (d), and tumor related forest (e)

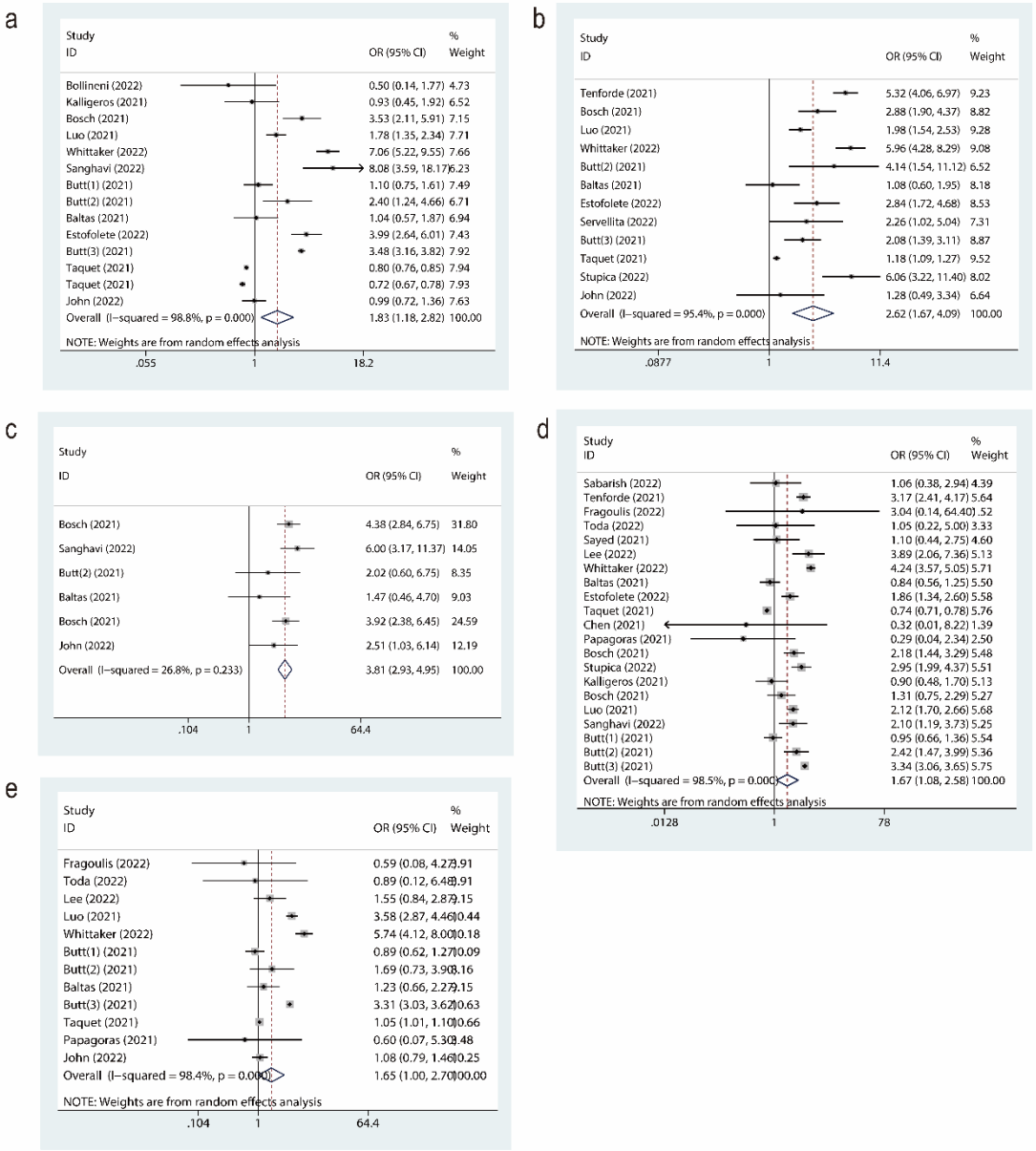


Figure S7. Liver disease related forest (a), lung disease related forest (b), hypertension related forest (c), and cerebrovascular disease related forest (d)

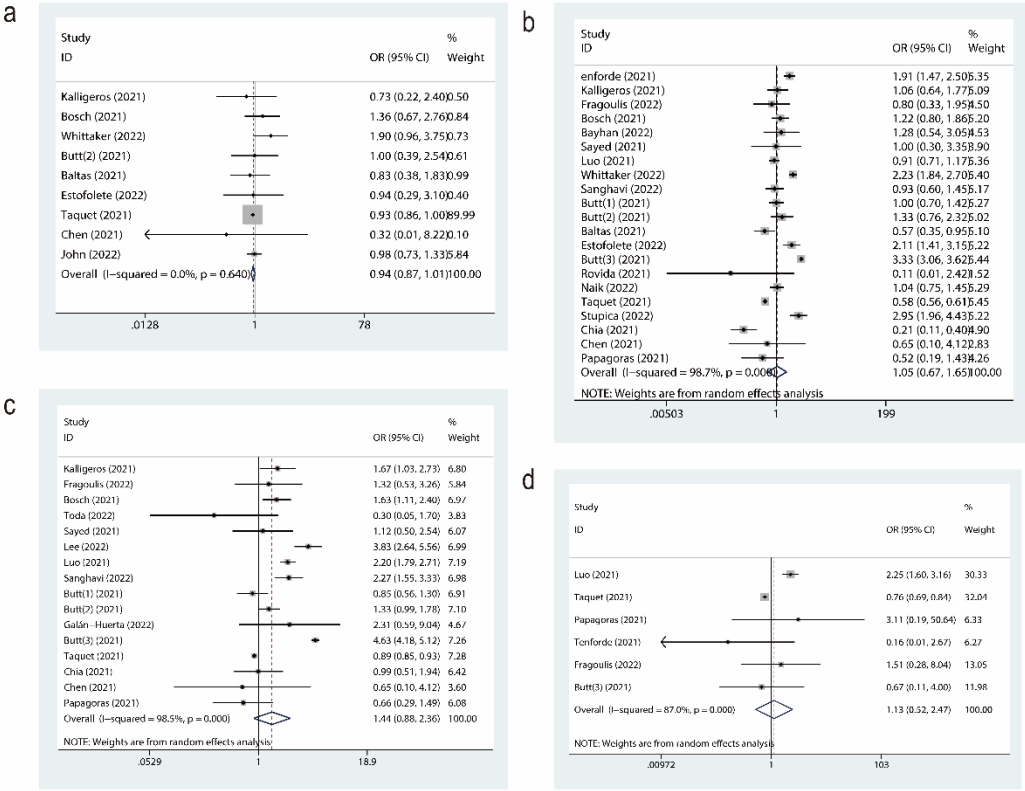


Figure S8. Fever related forest (a), cough related forest (b), gastrointestinal symptoms related forest (c), dysgeusia/anosmia related forest (d), and neurological symptoms related forest (e)

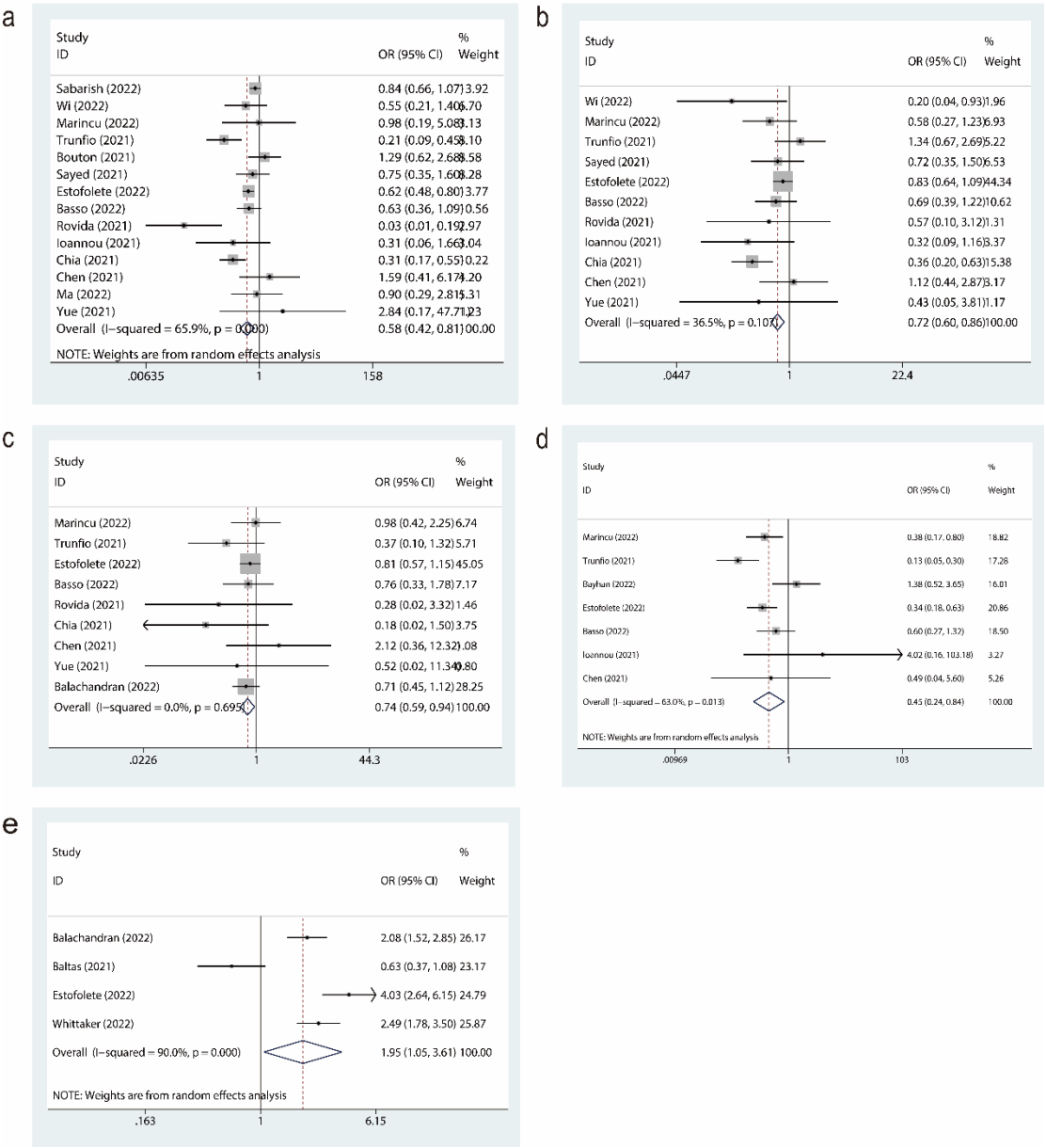


Figure S9. Anemia related forest (a), sore throat related forest (b), muscle and joint pain related forest (c), headache related forest (d), fatigue related forest (e), nausea/vomiting related forest (f), low oxygen saturation related forest (g), dyspnea related forest (h), and nasal congestion and runny nose related forest (i)

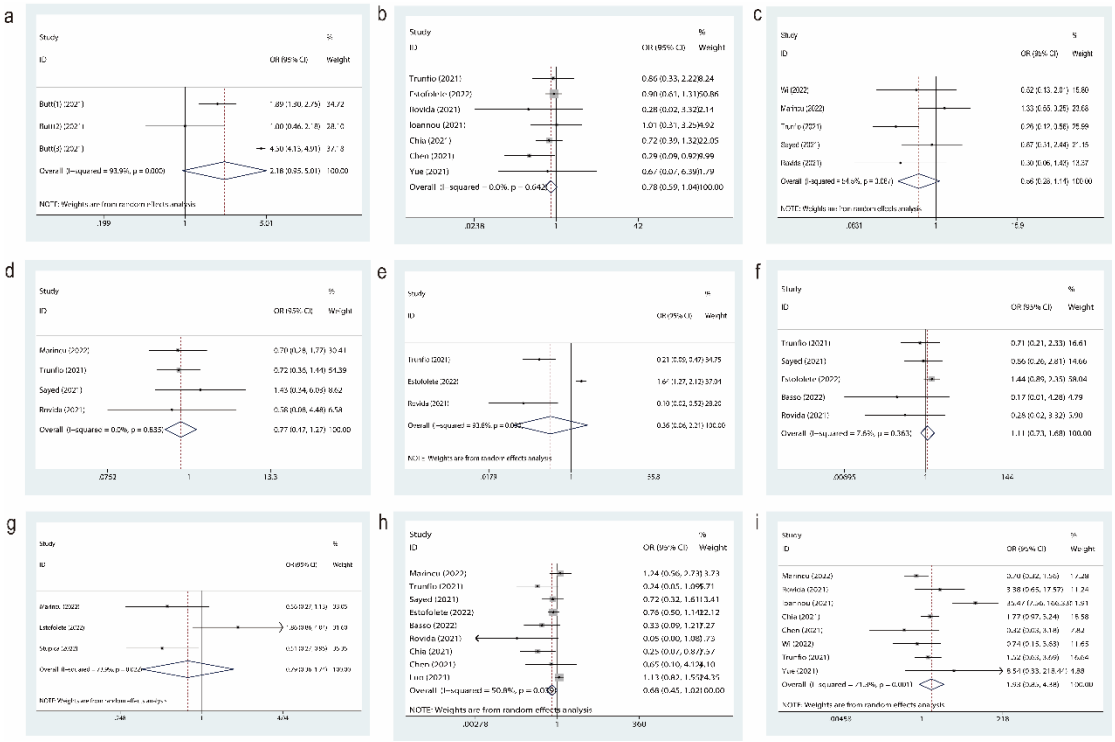
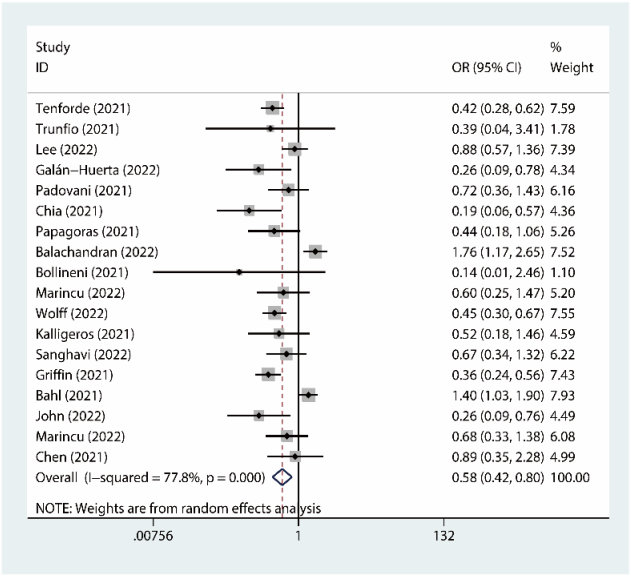
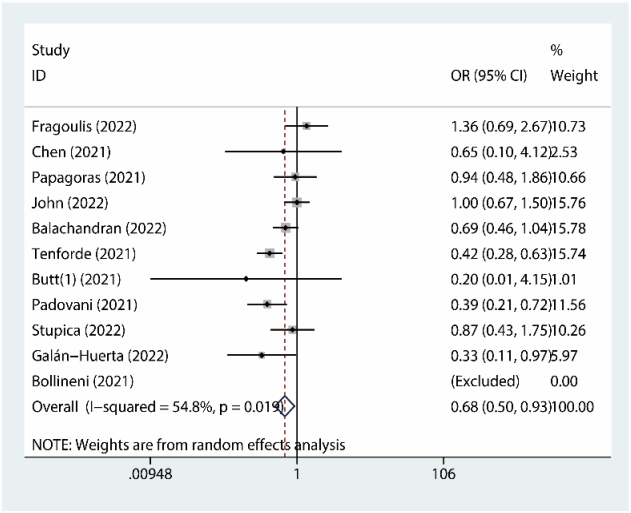


Figure S10. Oxygen support related forest (a), steroid related forest (b), and antiviral related forest (c)

a



b



c

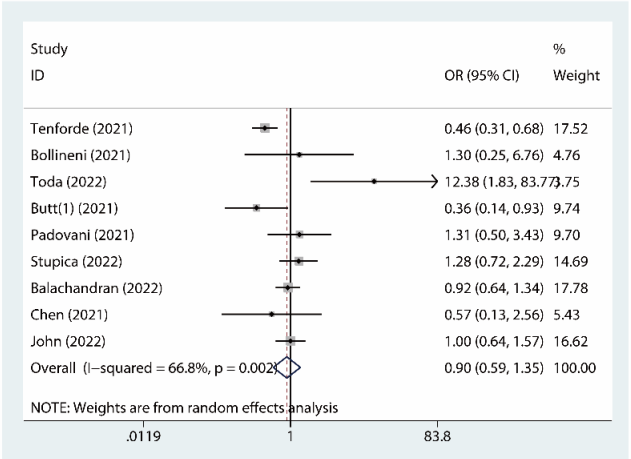
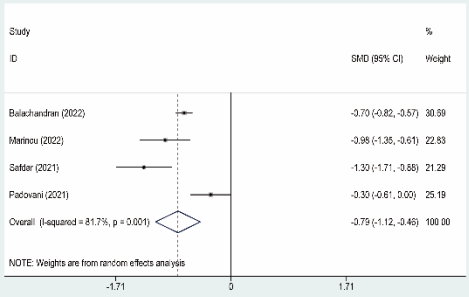
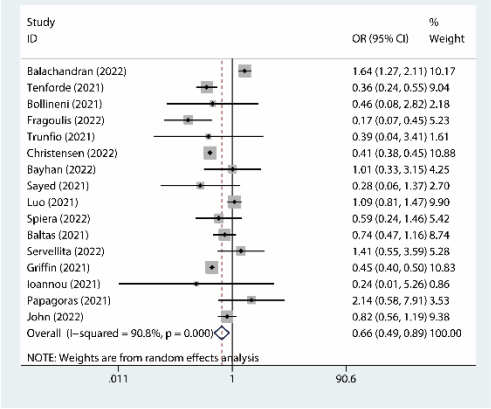


Figure S11. Forest plots related to hospitalization

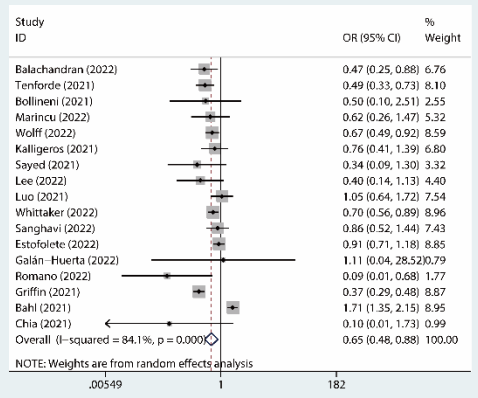
a


Study
ID
SMD (95% CI)
%
Weight
Balachandran (2022) -0.70 (-0.82, -0.57) 30.59
Marincu (2022) -0.98 (-1.35, -0.61) 22.83
Safdar (2021) -1.30 (-1.71, -0.88) 21.29
Padovani (2021) -0.30 (-0.61, 0.00) 25.19
Overall (I-squared = 81.7%, p = 0.001) -0.79 (-1.12, -0.46) 100.00
NOTE: Weights are from random effects analysis
-1.71
0
1.71

b


Study
ID
OR (95% CI)
%
Weight
Balachandran (2022) 1.64 (1.27, 2.11) 10.17
Tenforde (2021) 0.36 (0.24, 0.55) 9.04
Bollineni (2021) 0.46 (0.08, 2.82) 2.18
Fragoulis (2022) 0.17 (0.07, 0.45) 5.23
Trunfio (2021) 0.39 (0.04, 3.41) 1.61
Christensen (2022) 0.41 (0.38, 0.45) 10.88
Bayhan (2022) 1.01 (0.33, 3.15) 4.25
Sayed (2021) 0.28 (0.06, 1.37) 2.70
Luo (2021) 1.09 (0.81, 1.47) 9.90
Spiera (2022) 0.59 (0.24, 1.46) 5.42
Baltas (2021) 0.74 (0.47, 1.16) 8.74
Servellita (2022) 1.41 (0.55, 3.59) 5.28
Griffin (2021) 0.45 (0.40, 0.50) 10.83
Ioannou (2021) 0.24 (0.01, 5.26) 0.86
Papagoras (2021) 2.14 (0.58, 7.91) 3.53
John (2022) 0.82 (0.56, 1.19) 9.38
Overall (I-squared = 90.8%, p = 0.000) 0.66 (0.49, 0.89) 100.00
NOTE: Weights are from random effects analysis
.011
1
90.6

c


Study
ID
OR (95% CI)
%
Weight
Balachandran (2022) 0.47 (0.25, 0.88) 6.76
Tenforde (2021) 0.49 (0.33, 0.73) 8.10
Bollineni (2021) 0.50 (0.10, 2.51) 2.55
Marincu (2022) 0.62 (0.26, 1.47) 5.32
Wolff (2022) 0.67 (0.49, 0.92) 8.59
Kalligeros (2021) 0.76 (0.41, 1.39) 6.80
Sayed (2021) 0.34 (0.09, 1.30) 3.32
Lee (2022) 0.40 (0.14, 1.13) 4.40
Luo (2021) 1.05 (0.64, 1.72) 7.54
Whittaker (2022) 0.70 (0.56, 0.89) 8.96
Sanghavi (2022) 0.86 (0.52, 1.44) 7.43
Estofolete (2022) 0.91 (0.71, 1.18) 8.85
Galán–Huerta (2022) 1.11 (0.04, 28.52) 0.79
Romano (2022) 0.09 (0.01, 0.68) 1.77
Griffin (2021) 0.37 (0.29, 0.48) 8.87
Bahl (2021) 1.71 (1.35, 2.15) 8.95
Chia (2021) 0.10 (0.01, 1.73) 0.99
Overall (I-squared = 84.1%, p = 0.000) 0.65 (0.48, 0.88) 100.00
NOTE: Weights are from random effects analysis
.00549
1
182

d

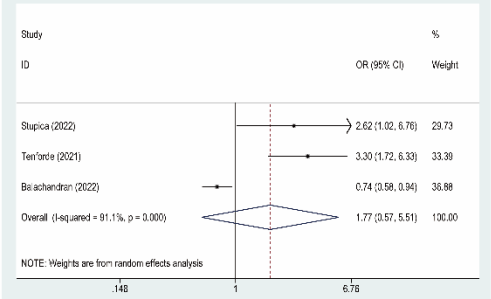
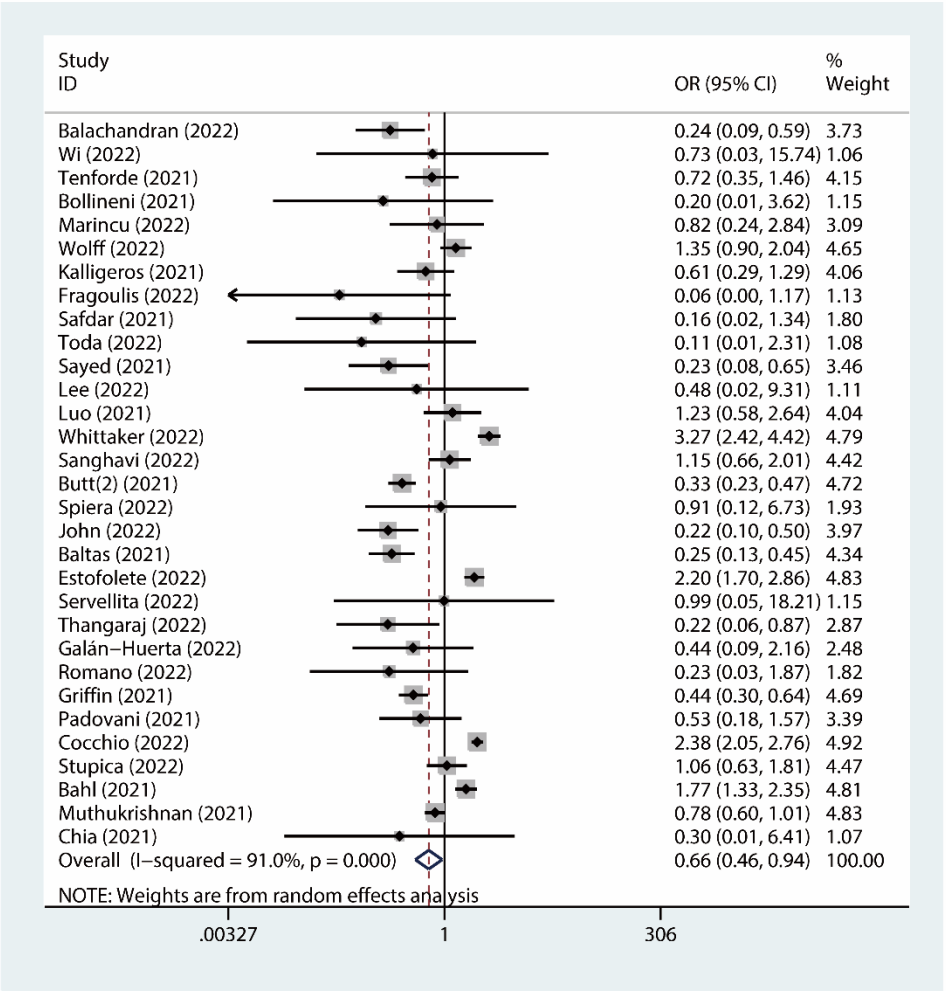

Study
ID
OR (95% CI)
%
Weight
Stupica (2022) 2.62 (1.02, 6.76) 29.73
Tenforde (2021) 3.30 (1.72, 6.33) 33.39
Balachandran (2022) 0.74 (0.58, 0.94) 36.88
Overall (I-squared = 91.1%, p = 0.000) 1.77 (0.57, 5.51) 100.00
NOTE: Weights are from random effects analysis
.148
1
6.76

Figure S12. Forest plots related to outcome and prognosis

a



b

